

Given

$$f_c = 1 \text{ kHz}$$

$$C = 10 \text{ nF}$$

$$\begin{aligned} R &= \frac{1}{2\pi f_c C} \\ &= \frac{1}{2\pi(1000)(10 \times 10^{-9})} \\ &= 15.9 \text{ k}\Omega \end{aligned}$$

Doubling R/C

$$\begin{aligned} f_c &= 500 \text{ Hz} \\ (R &= 31.8 \text{ kohm or } C = 20 \text{ nF}) \end{aligned}$$

Halving C/R

$$\begin{aligned} f_c &= 2 \text{ kHz} \\ (R &= 7.95 \text{ kohm or } C = 5 \text{ nF}) \end{aligned}$$